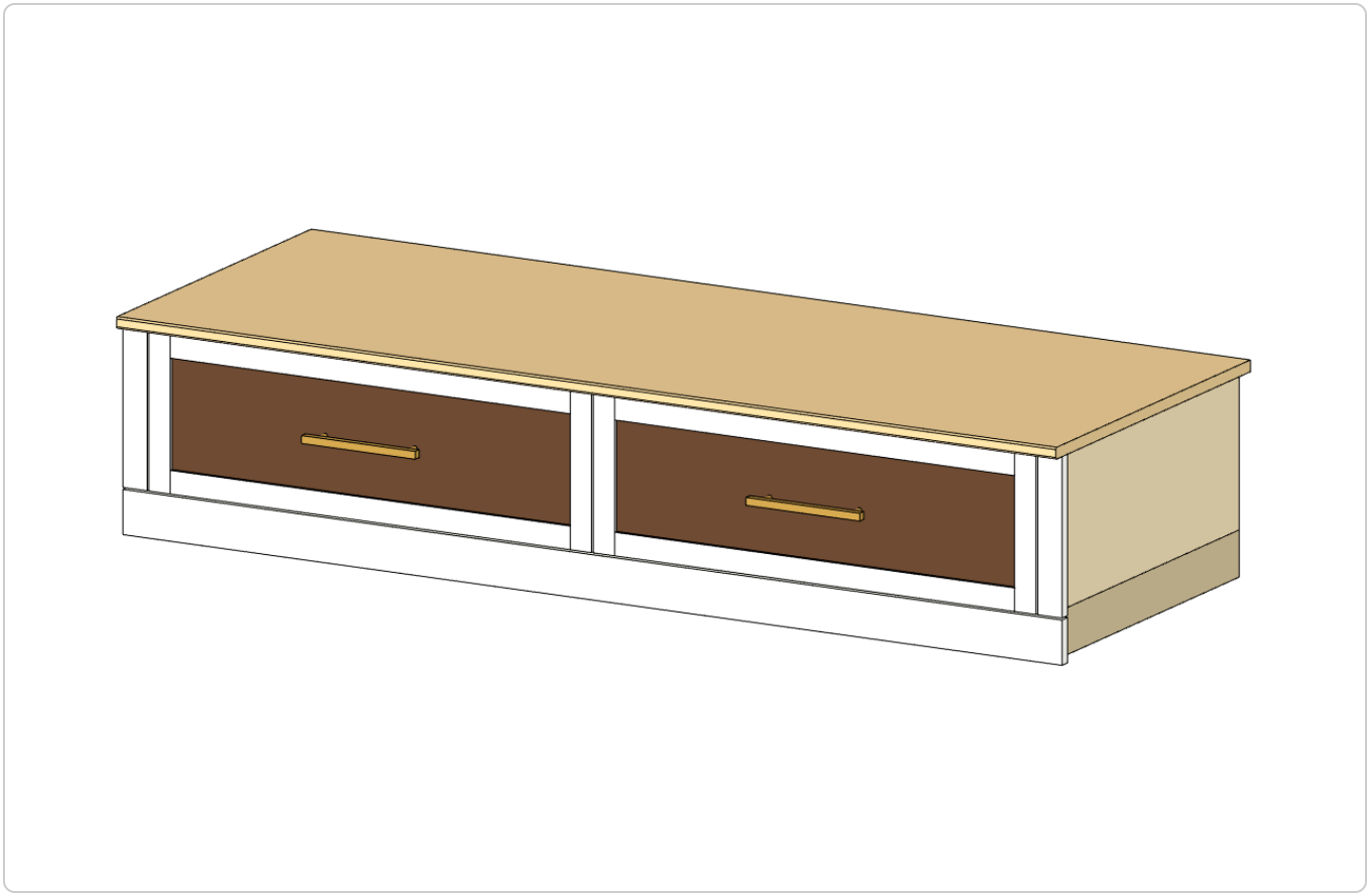


BUILT-IN BENCH · 66 X 28 IN ALCOVE

Built-in Bench Cut List

Two drawers on Blum undermounts, white-painted maple frames around inset walnut panels, solid maple top. 35 pieces: the case first, then the drawers, each grouped by material in cutting order. Inches to the nearest 1/32. Every part has a shop drawing with its grooves, rabbets, notches and bores located from the blank's own edges.



Measure the alcove at the floor, 16 in and 36 in, at the back wall and the pilaster faces, before cutting. Cut the top, plinth and scribe strips long and scribe them. Confirm the low receptacle seen in the photo before cutting the back.

Case 15 pieces

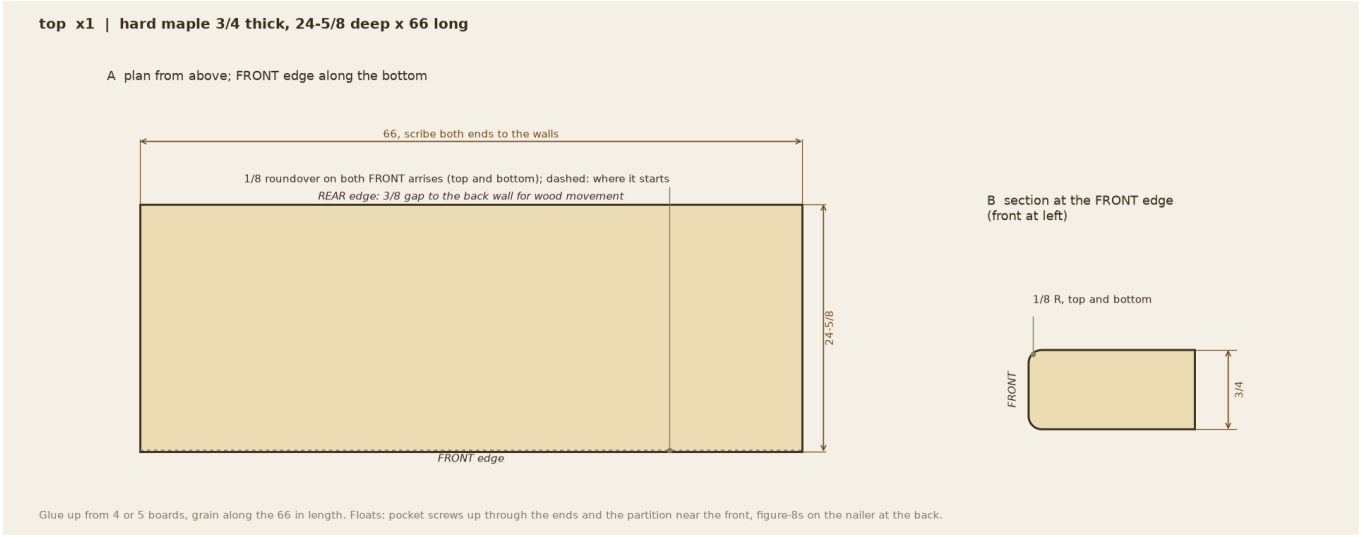
Hard maple: top (bare), nailer; plinth and scribe strips (painted white)

5 pieces · 13.9 sq ft face, no kerf or waste

1× top

3/4 x 24-5/8 x 66

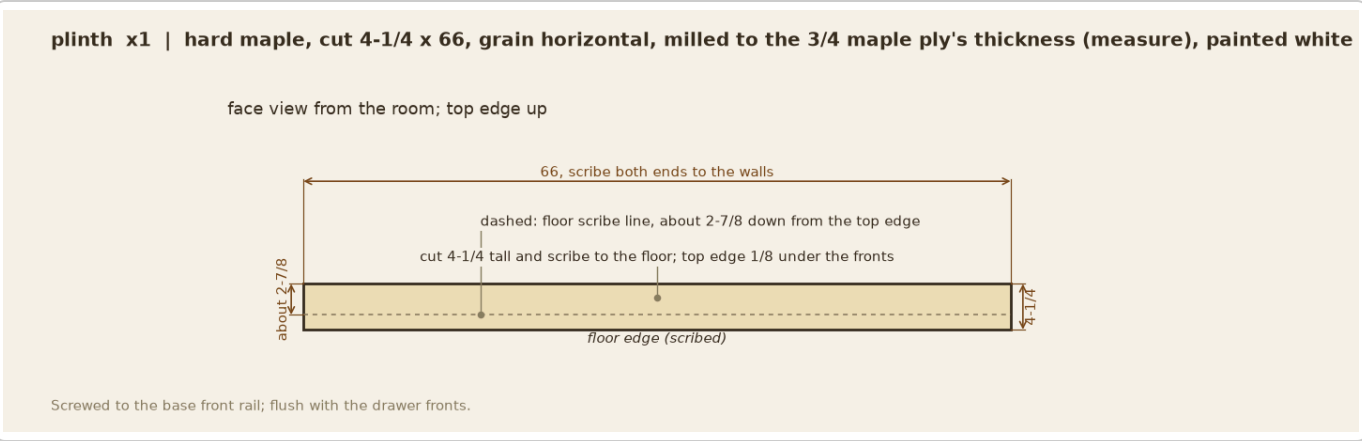
glue up from 4 or 5 boards, grain along the 66 in; 1/8 roundover on both front arrises; scribe the ends to the walls; 3/8 gap at the back wall; pocket screws up through the ends and partition near the front, figure-8s on the nailer at the back



1× plinth

23/32 x 2-7/8 x 66

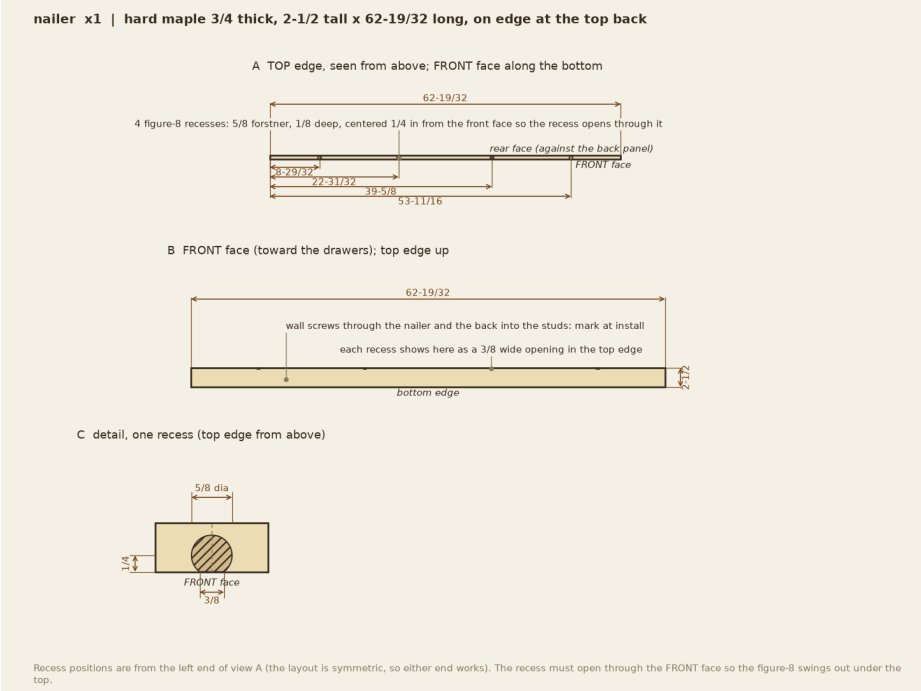
cut 3-1/4 tall and scribe to the floor so the top edge sits 1/8 under the fronts; scribe the ends to the walls; grain horizontal; mill to the ply thickness; screwed to the base front rail; painted white



1× **nailer**

3/4 x 2-1/2 x 62-19/32

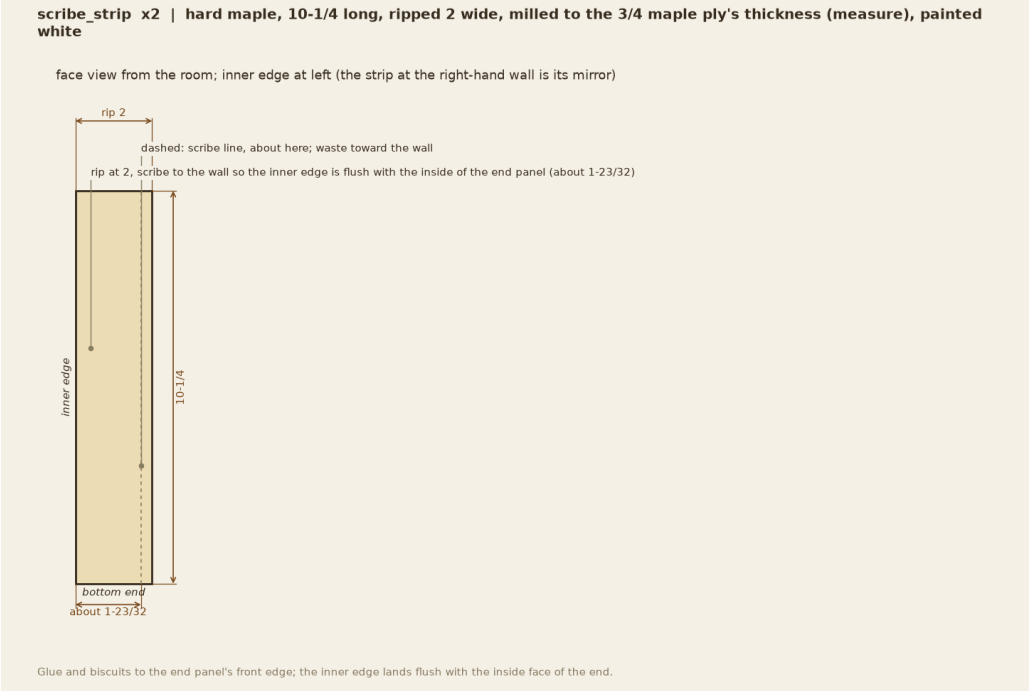
on edge against the back panel, through the partition notch; four 5/8 figure-8 recesses 1/8 deep on the top edge, centered 1/4 in from the front face; screw through it and the back into the wall studs



2× scribe_strip

23/32 x 1-23/32 x 10-1/4

rip at 2, scribe to the wall so the inner edge lands flush with the inside of the end panel (about 1-23/32); glue and biscuits to the end's front edge; mill to the ply thickness; painted white



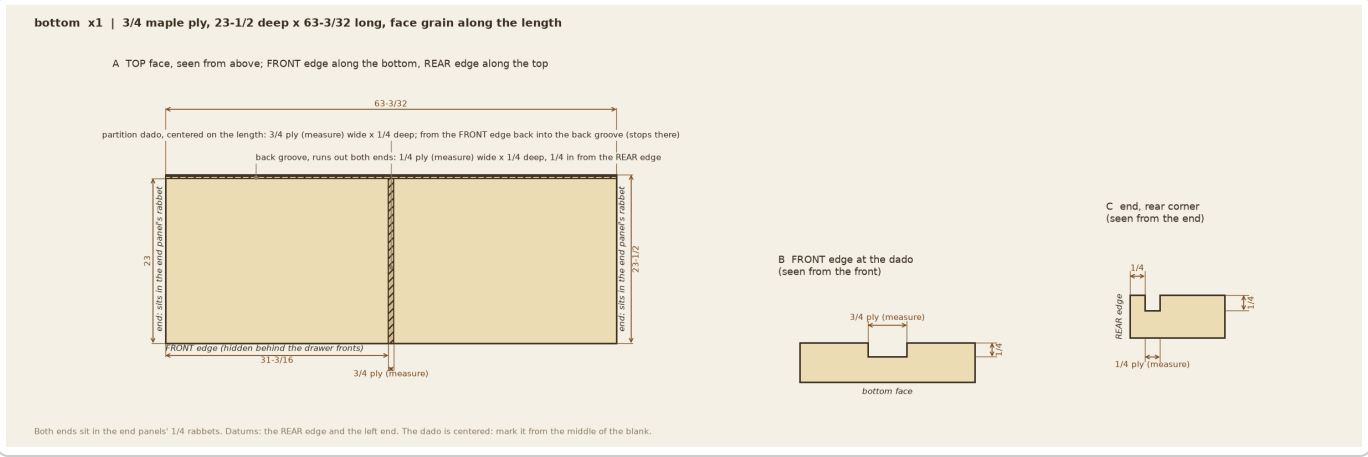
3/4 in maple plywood (measure the sheet): ends, bottom, partition

4 pieces · 15.2 sq ft face, no kerf or waste

1× bottom

23/32 x 23-1/2 x 63-3/32

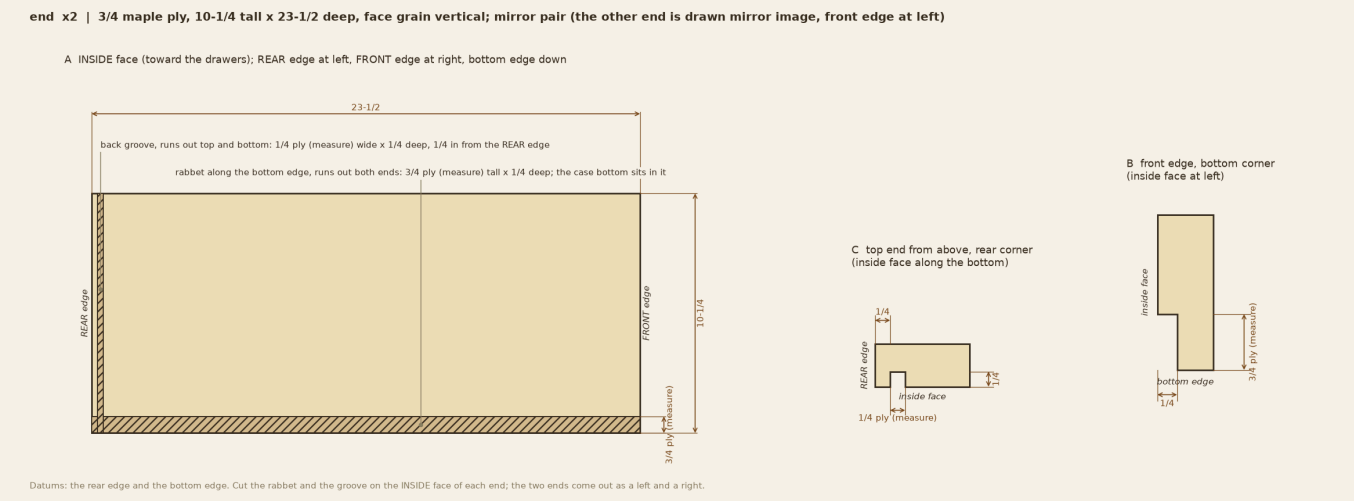
face grain across the 64 in; 1/4 deep back groove 1/4 from the rear edge; 1/4 deep x 3/4 ply (measure) dado for the partition, centered, from the back groove to the front edge; front edge is covered by the drawer fronts



2× **end** see drawing: not a plain rectangle

23/32 x 10-1/4 x 23-1/2

face grain vertical; 1/4 x 23/32 (measure) rabbet along the bottom inside edge for the case bottom; 1/4 deep x 1/4 ply (measure) groove 1/4 from the rear edge, through; mirror pair



1× **partition**

23/32 x 9-25/32 x 23

face grain vertical; sits in the bottom dado and against the back panel; notch the top rear corner 3/4 deep x 2-1/2 tall for the nailer



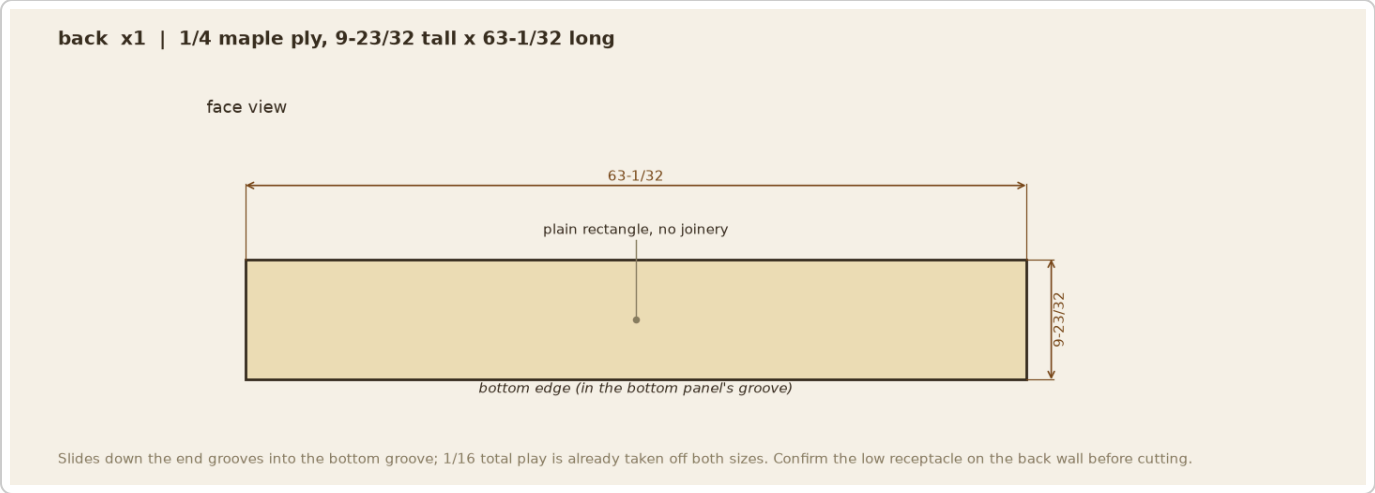
1/4 in maple plywood (measure the sheet): back

1 pieces · 4.3 sq ft face, no kerf or waste

1× **back**

1/4 x 9-23/32 x 63-1/32

plain rectangle; slides down the end grooves into the bottom groove; confirm the low receptacle on the back wall before cutting



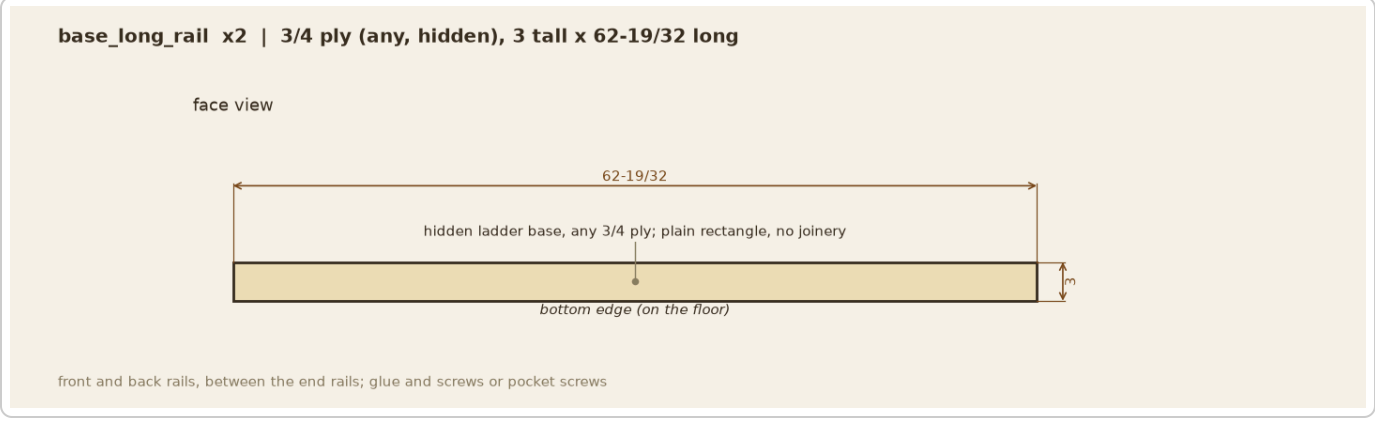
3/4 in plywood, any grade (hidden): base ladder

5 pieces · 4.0 sq ft face, no kerf or waste

2× **base_long_rail**

23/32 x 3 x 62-19/32

front and back rails between the end rails; pocket screws or glue and screws



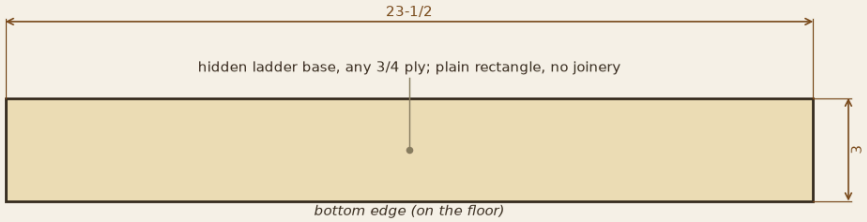
2× base_end_rail

23/32 x 3 x 23-1/2

ladder base, 3 tall; the end panels stand on these

base_end_rail x2 | 3/4 ply (any, hidden), 3 tall x 23-1/2 long

face view



front to back under each end panel; the end panels stand on these

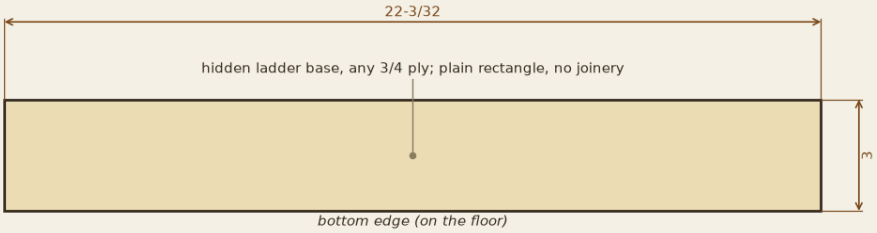
1× base_center_rail

23/32 x 3 x 22-3/32

under the partition

base_center_rail x1 | 3/4 ply (any, hidden), 3 tall x 22-3/32 long

face view



between the long rails, under the partition

Drawers 20 pieces

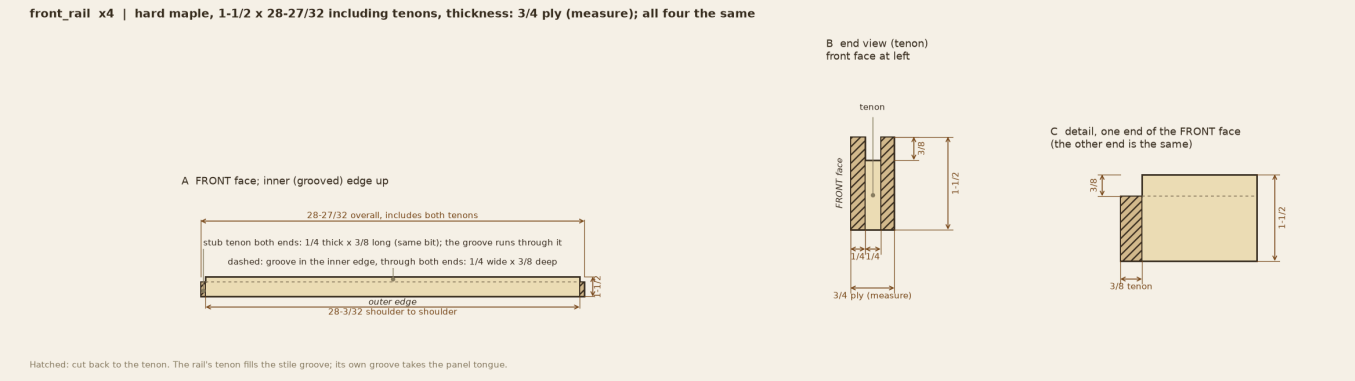
Drawer front frames, hard maple, painted white: stiles and rails

8 pieces · 1.6 sq ft face, no kerf or waste

4× front_rail see drawing: not a plain rectangle

23/32 x 1-1/2 x 28-27/32

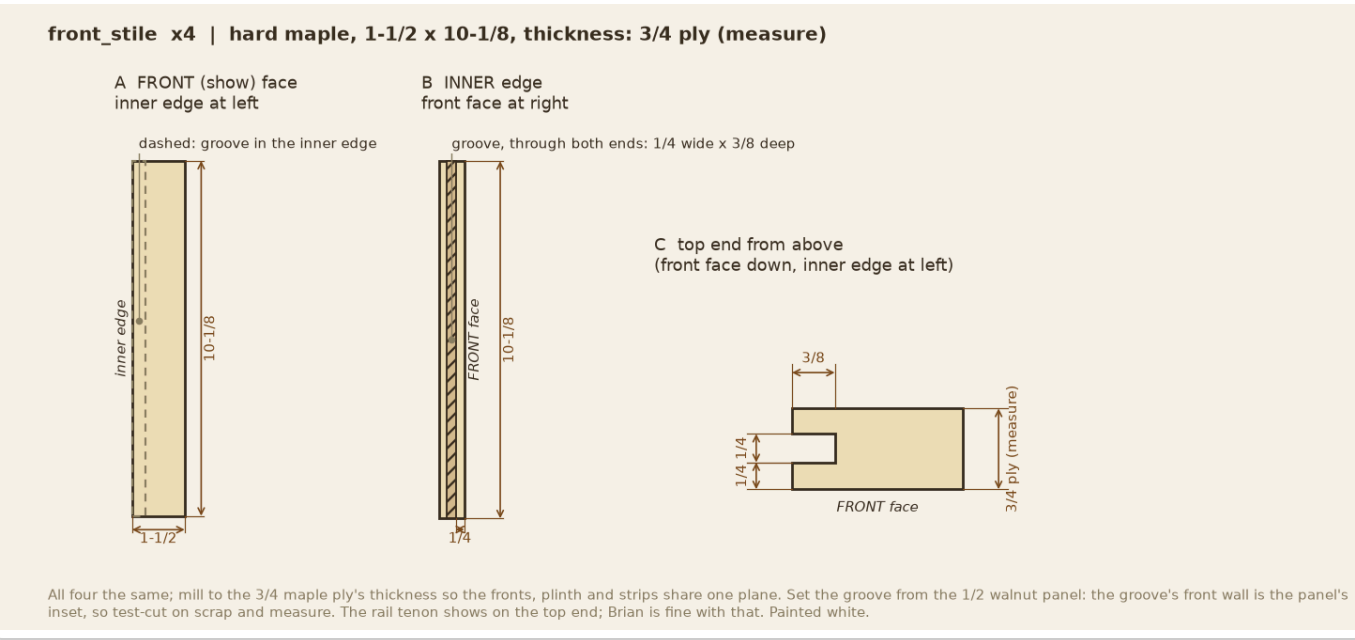
1/4 x 3/8 groove on the inner edge, through; 1/4 x 3/8 stub tenon on both ends (same bit); length includes both tenons; painted white



4× front_stile see drawing: not a plain rectangle

23/32 x 1-1/2 x 10-1/8

1/4 x 3/8 groove on the inner edge, through, front wall 1/4 thick (set the fence from the panel ply); the rail tenons show on both ends (the top one is the visible one); painted white



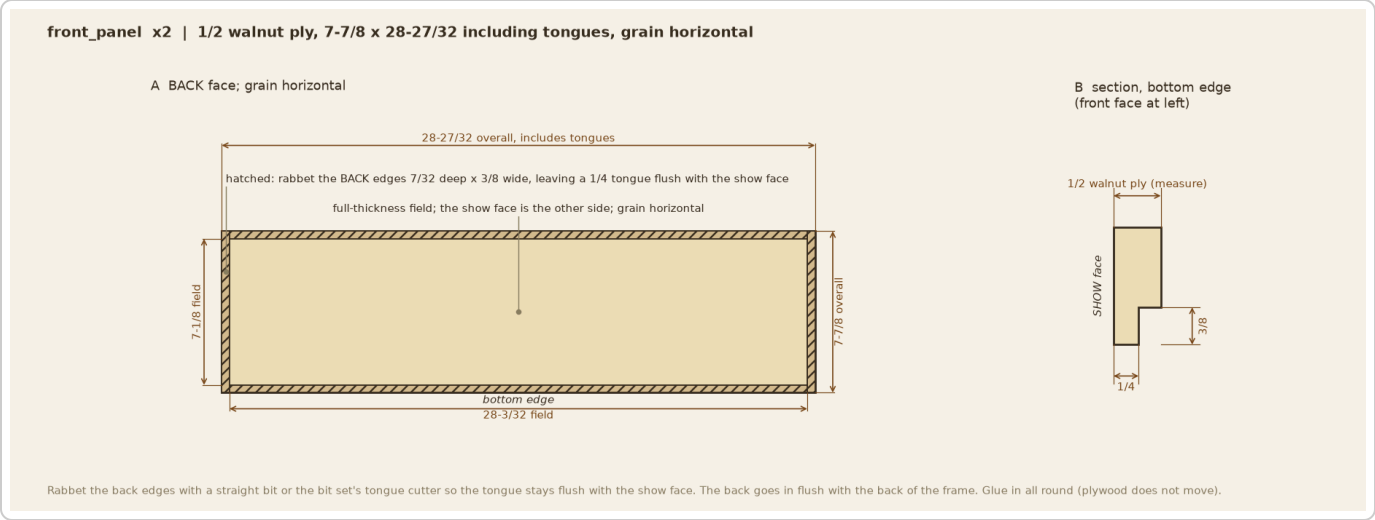
Drawer front panels, 1/2 in walnut plywood, left bare (measure the sheet)

2 pieces · 3.2 sq ft face, no kerf or waste

2× `front_panel` see drawing: not a plain rectangle

15/32 x 7-7/8 x 28-27/32

1/4 x 3/8 tongue on all four edges, flush with the show face (rabbet the BACK edges 7/32 deep); grain horizontal; size includes the tongues; back flush with the frame back, face 1/4 below the maple; glue in (plywood, no movement)



Drawer boxes: sides, fronts, backs and bottoms, 1/2 in 9-ply Baltic birch (measure the sheet)

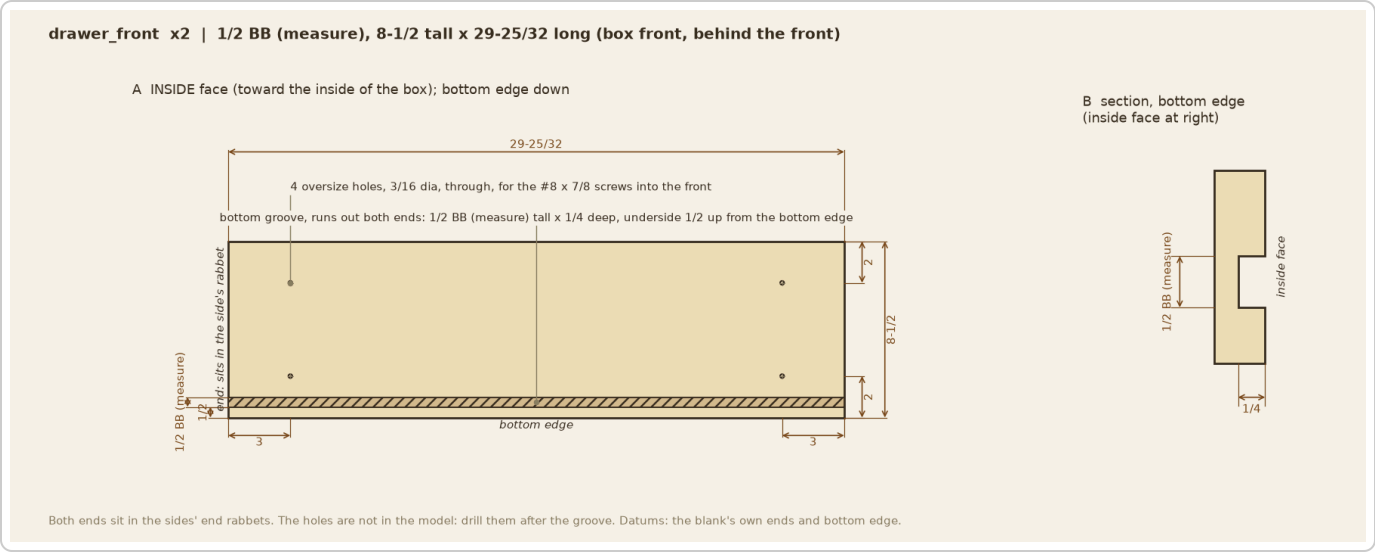
10 pieces · 20.4 sq ft face, no kerf or waste

2× drawer_front

see drawing: not a plain rectangle

15/32 x 8-1/2 x 29-25/32

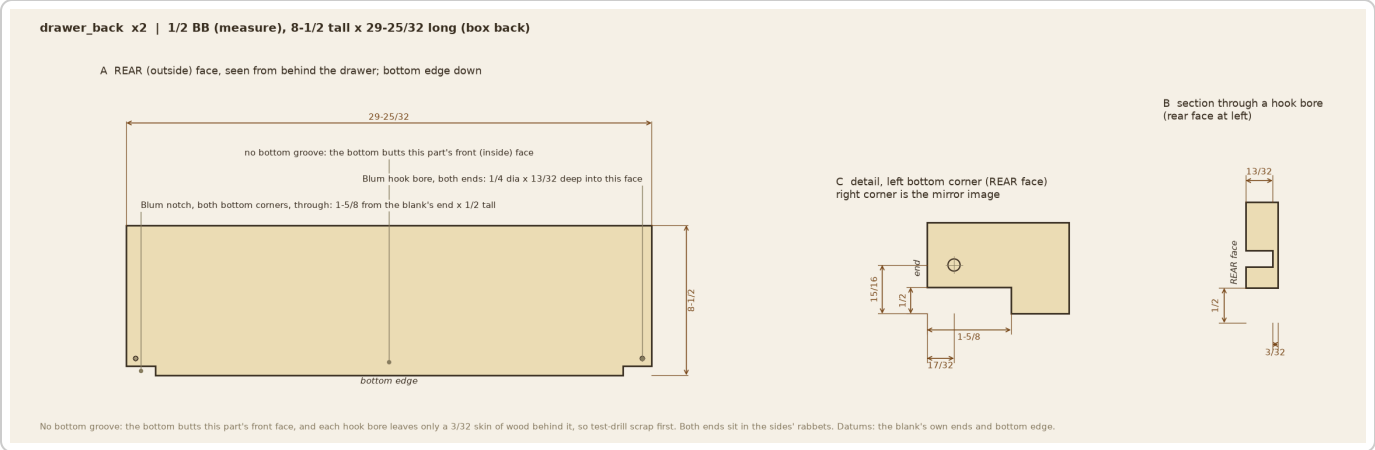
box front, sits in the side rabbets; 1/4 deep bottom groove to match the sides; the front screws to it from inside through 4 oversize holes: #8 x 7/8 in, pre-drilled, so the bite stays short of the face; the box front overlaps the top rail by 1-1/8 in (best bite) and the panel field, but the bottom rail by only 1/4 in



2× drawer_back

15/32 x 8-1/2 x 29-25/32

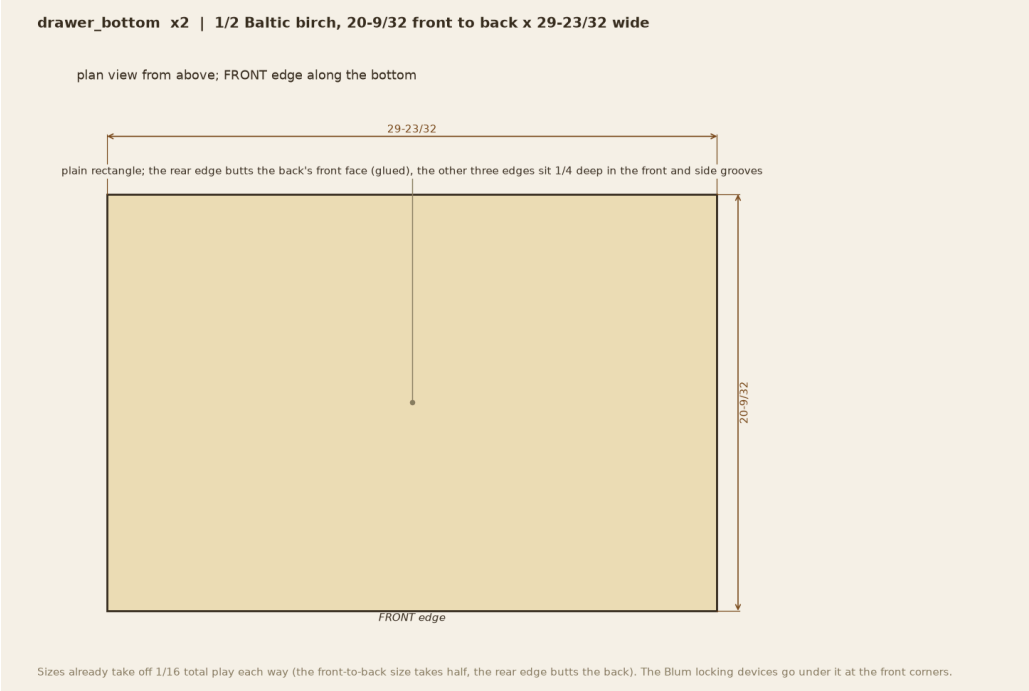
box back, sits in the side rabbets; no bottom groove (a groove would break into the hook bores), the bottom butts its front face; Blum notch 1-3/8 in from each side's inner face x 1/2 tall at both bottom corners; hook bores 1/4 dia x 13/32 deep into the rear face, centered 9/32 in from each side's inner face and 15/16 above the bottom edge (Blum 563H sheet)



2× drawer_bottom

15/32 x 20-9/32 x 29-23/32

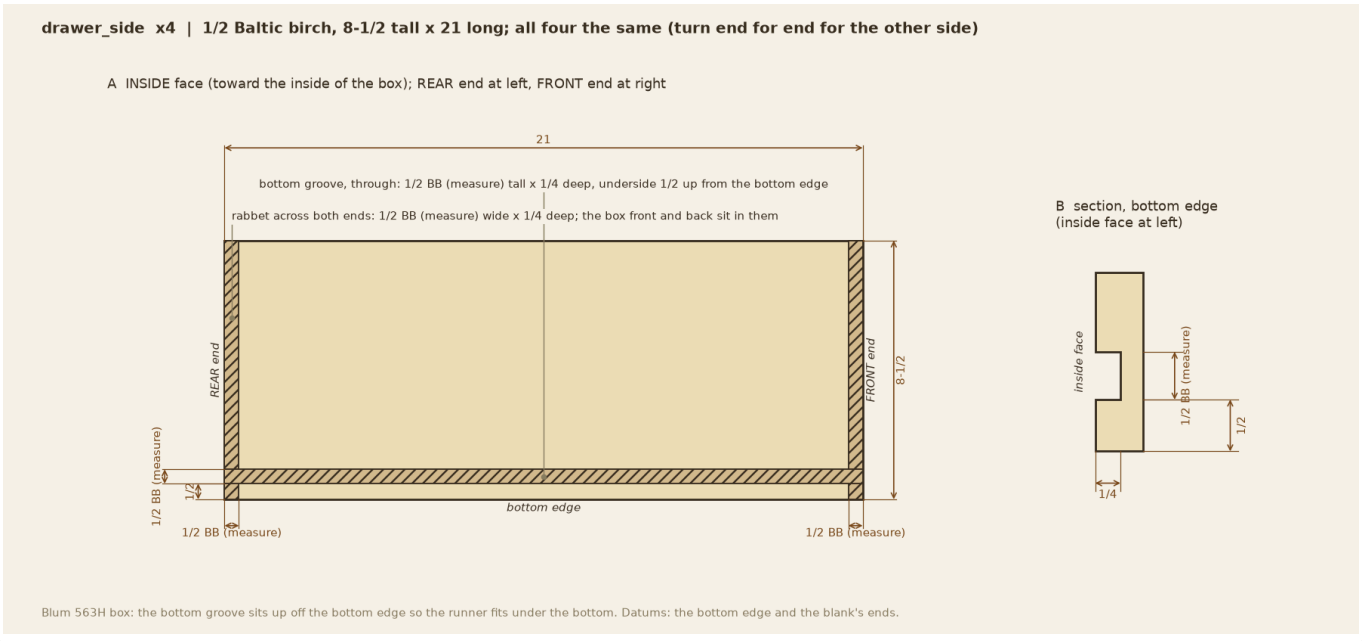
plain rectangle: 1/4 into the side and front grooves, butted to the back and glued to it; the Blum locking devices sit flush under it at the front corners (their screws go into the box front)



4× drawer_side see drawing: not a plain rectangle

15/32 x 8-1/2 x 21

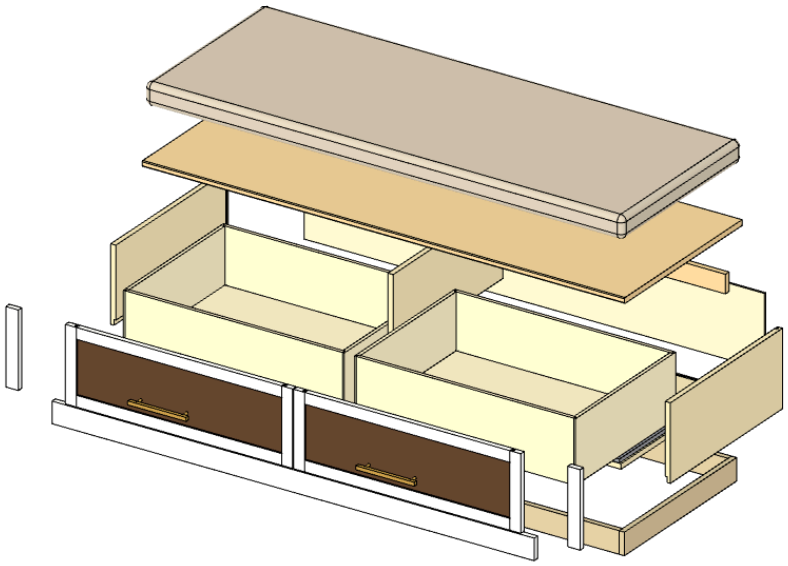
1/4 deep x 1/2 BB (measure) rabbet across both ends on the inside face; 1/4 deep x 1/2 BB (measure) bottom groove, underside 1/2 up from the bottom edge, through



Hardware from the design; not in the cut list

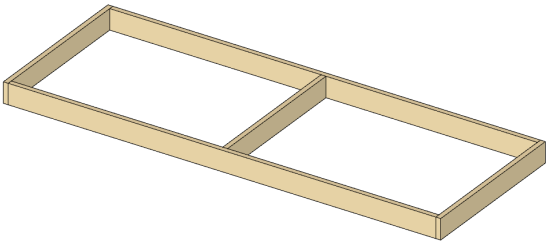
2 pair	**Blum TANDEM plus BLUMOTION 563H, 21 in (563H5330B)**	one pair per drawer; 90 lb dynamic; runner front 3/32 behind the case front edge; no rear brackets in a frameless case
2 pair	**Blum locking devices T51.1901 R and L**	front corners under the drawer bottom, screwed into the 1/2 box front with #6 x 1/2
2	**Brass bar pull, 8 in**	centered on each front; break-away machine screws through the 1/2 panel plus the 1/2 box front
4	**Figure-8 tabletop fasteners**	nailer to top; recess centered 1/4 in from the front face so it opens through
6	**Pocket screws #8 x 1-1/4 fine**	ends and partition to the top, near the front
8	**Wood screws #8 x 7/8**	fronts to the box fronts, through oversize holes, pre-drilled
1	**Bench cushion**	about 65-1/2 x 25 x 3, boxed, full depth to the wall

Assembly 11 steps, in order



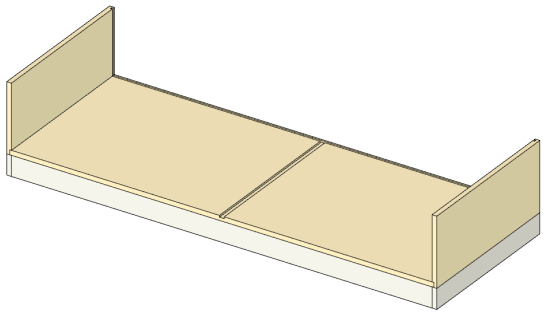
1. Base ladder

Glue and screw the five rails into a 64 x 23-1/2 ladder, 3 in tall. Set it in the alcove with its front edge 23-1/2 in from the back wall, level it on shims and screw it to the floor.



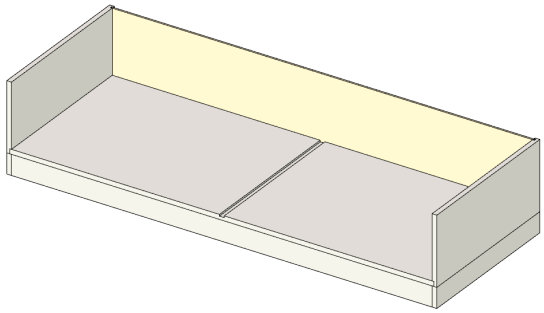
2. Ends and bottom

Stand the two end panels on the base end rails, rabbets and grooves facing in, grooves at the rear. Glue the case bottom into the rabbets along the bottom of both ends and screw up through the base. Check for square.



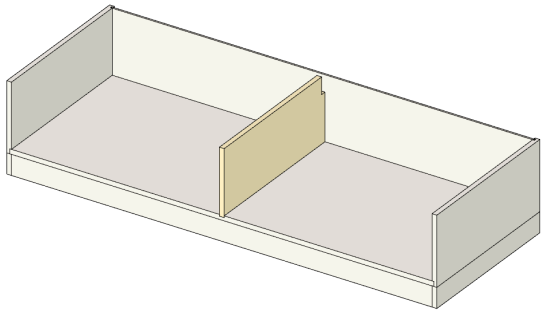
3. Back

Slide the 1/4 back down the end grooves until it seats in the bottom groove. No glue; the nailer pins it later.



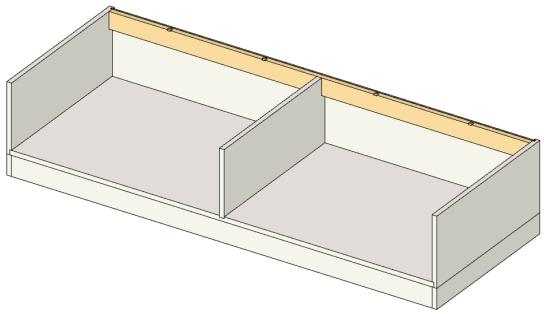
4. Partition

Drop the partition into the dado in the bottom with its notch at the top rear, front edge flush with the ends. Screw up through the bottom.



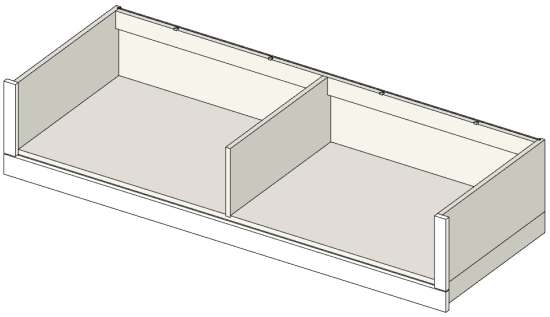
5. Nailer

Slide the nailer in on edge through the partition notch, tight against the back and flush with the panel tops. Screw through the ends into its ends, then through the nailer and the back into the wall studs.



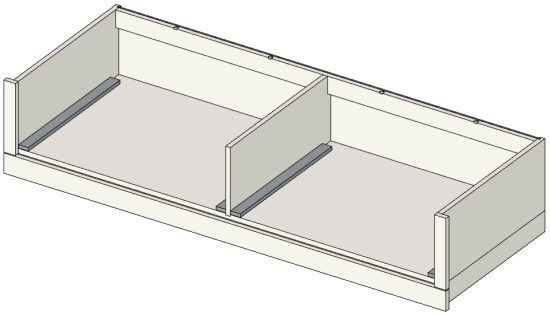
6. Scribe strips and plinth

Scribe both strips to the walls and glue them to the ends' front edges with biscuits, inner edges flush with the inside of the ends. Scribe the plinth to the floor and screw it to the base front rail with its top edge 1/8 below the case bottom. Paint the strips and plinth white.



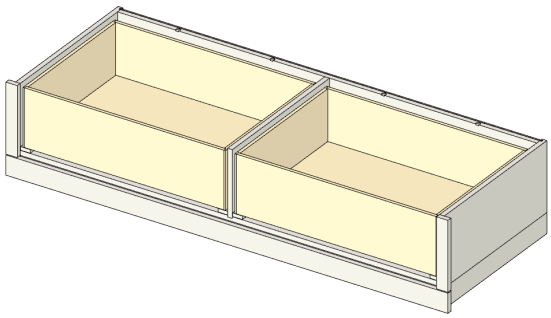
7. Slides

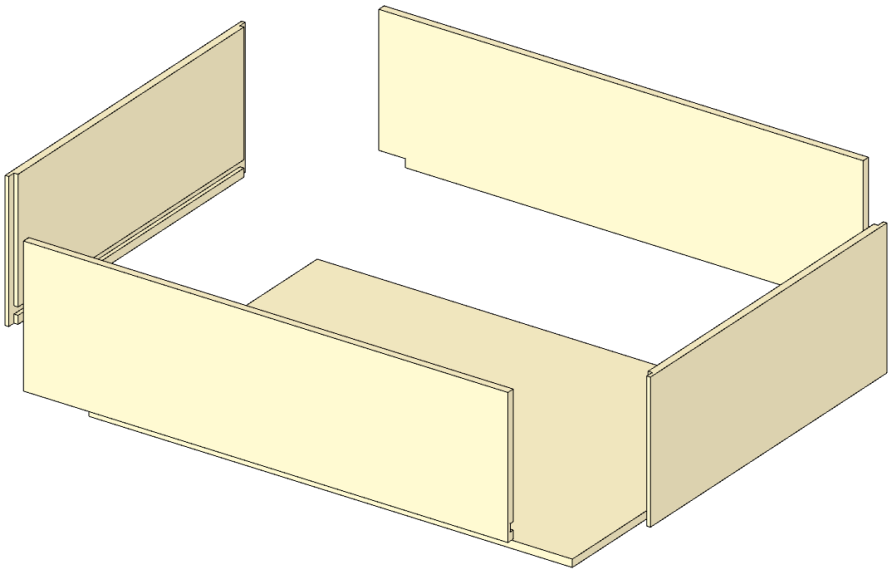
Screw the Blum cabinet members to the ends and the partition, sitting on the case bottom, fronts 3/32 behind the case front edge. Use the five marked holes per runner.



8. Drawer boxes

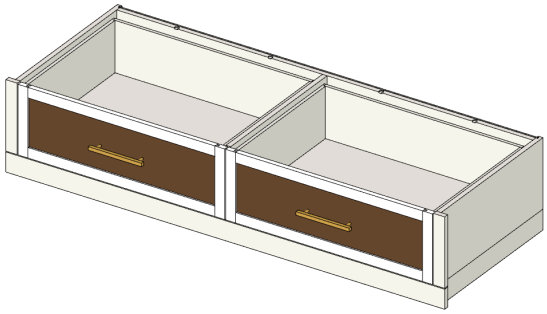
Glue each 1/2 birch box up around its bottom: front and back in the side rabbets, bottom in the side and front grooves with its underside 1/2 above the side edges, butted to the back and glued to it. Notches and hook bores at the rear (drill scrap first: the bore leaves a thin skin). Clip the locking devices under the bottom at the front corners with #6 x 1/2 screws and set the boxes on the runners.

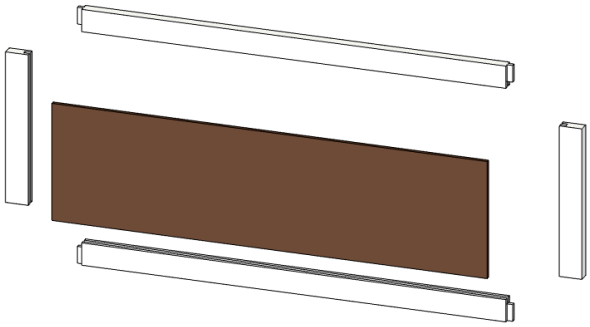




9. Fronts and pulls

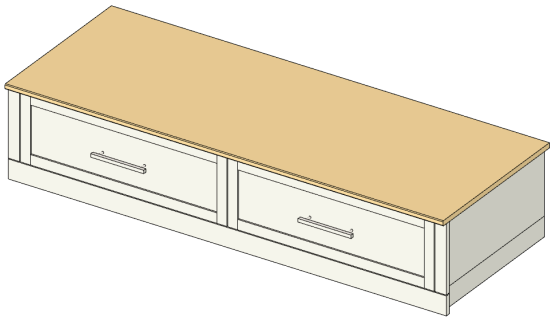
Glue up each 1/2 walnut panel in its maple frame, back faces flush so the panel sits 1/4 below the maple (groove every piece first, then cut the tongues and tenons to fit). Prime and paint the frames white. Hang the fronts with 1/8 reveals all round, screwed from inside the box through oversize holes. Center the pulls.





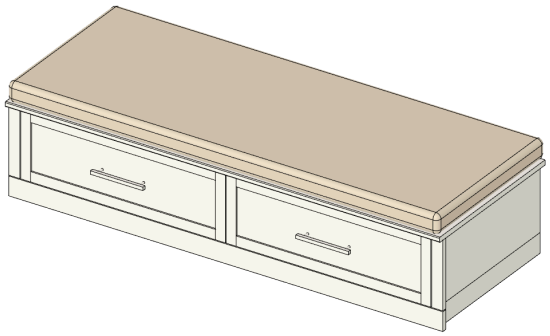
10. Top

Set the top with its front edge 25 in from the back wall (3 in back from the pilaster faces) and 3/8 off the back wall. Pocket screws up into it through the ends and partition near the front; figure-8s on the nailer at the back so it can move toward the wall.



11. Cushion

Cushion on. Done.



Generated 2026-10-23 from `projects/Built-in-bench/cutlist.csv` and `images/parts/`, which `built_in_bench.py` and `part_drawings.py` write on every export; every drawn cut is probed against the CAD solid before its sheet is written. The bench image at the top and the assembly pictures are screenshots of the CAD viewer (`hero_shot.py`, `assembly_shots.py`). Part names match the CAD viewer.